

MACHINE LEARNING • REGRESSION PROJECT

Predicting Gas Turbine CO Emissions from Sensor Data

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7,411
records

11
variables

R^2 0.83
best model

Project Overview

Why predict CO, and how

The project

Predict carbon monoxide (CO) emitted by a gas turbine using ambient and process sensor readings.

Why it matters

CO is a regulated pollutant. Prediction supports emission monitoring and cleaner operation.

Approach

Supervised learning: regression on a continuous target, comparing 5 models.

Objective

Find the model that predicts CO most accurately on unseen data.

Dataset & Target

Gas turbine sensor and emission records

7,411

records

11

columns

0

nulls / duplicates

Ambient

AT • AP • AH

Turbine process

AFDP • GTEP • TIT • TAT • TEY • CDP

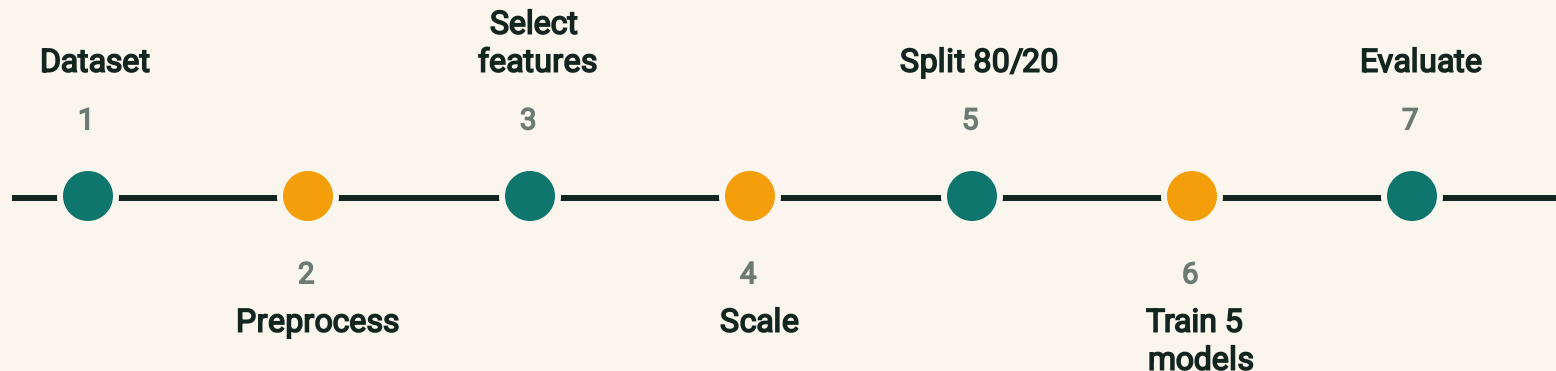
Emissions

CO (target) • NOX

Target: CO — a continuous value from 0 to 43.6 (mean 1.57)

Machine Learning Workflow

From raw sensor data to model comparison



Clean the data, keep the strongest features, then let five regressors compete.

Data Preprocessing

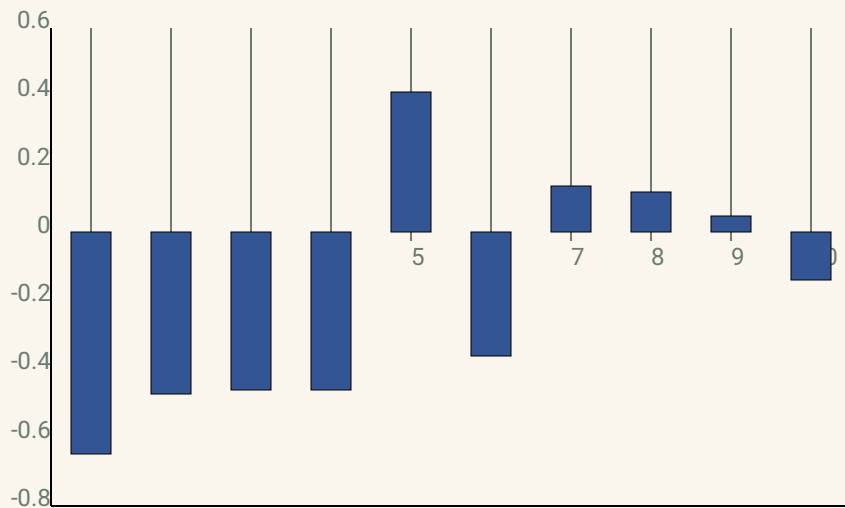
Five checks before modeling

1	Inspect	Shape, columns, types, summary statistics
2	Nulls	None found in any column
3	Duplicates	0 duplicate rows
4	Correlation	Heatmap shows strong turbine-variable links
5	Outliers	IQR capping ($Q1 - 1.5 \cdot IQR$, $Q3 + 1.5 \cdot IQR$)

Purpose: clean, reliable inputs for every model.

What Drives CO?

Correlation of each variable with CO



Strongest link

TIT (turbine inlet temperature): $r = -0.65$

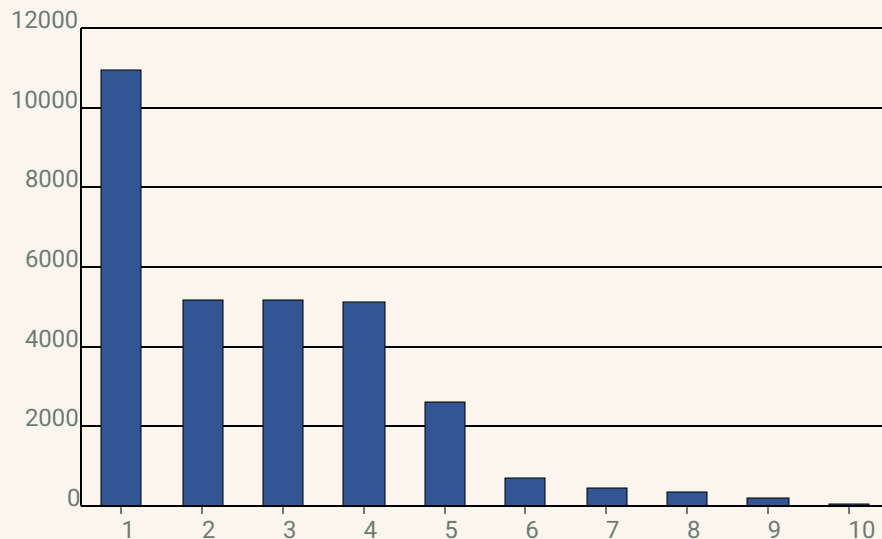
Reading

Hotter, higher-load operation goes with lower CO. Ambient factors matter less.

Turbine variables, not weather, carry most of the signal.

Feature Selection & Scaling

SelectKBest (f_regression), then StandardScaler



F-scores rank TIT first by a wide margin.

10 features kept

5,928 train rows

1,483 test rows

Regression Models

Five algorithms, one test set

1

Linear Regression

Straight-line baseline

2

Decision Tree

Splits data by rules

3

Random Forest

Averages many trees

4

AdaBoost

Learners fix earlier errors

5

Gradient Boosting

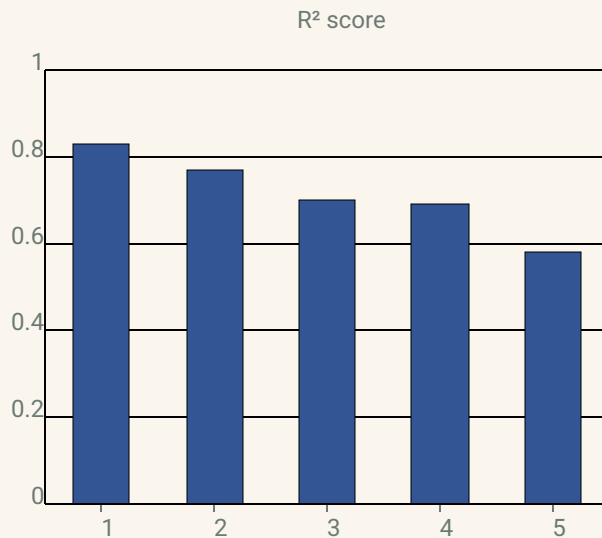
Minimizes loss step by step

Metrics: MAE, MSE and R^2 on the held-out test set.

Model Results

Random Forest leads on every metric

Model	MAE	MSE	R ²
Random Forest	0.23	0.13	0.83
Gradient Boosting	0.29	0.17	0.77
Linear Regression	0.35	0.23	0.70
Decision Tree	0.31	0.24	0.69
AdaBoost	0.47	0.32	0.58



Random Forest: R² 0.83, MAE 0.23, MSE 0.13 — best of the five.

Conclusion

Random Forest is the best model (R^2 0.83)

TIT is the most influential predictor of CO

Ensembles beat the linear baseline (R^2 0.70), so relationships are non-linear

Thank You